

STATS 20: Discussion 2 – Chapter 1 & Sections 2.1-2.2

STATS 20 Team

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Today's Goal

For this discussion section, you are tasked to create your *own* RMarkdown file and make sure you can answer questions concerning

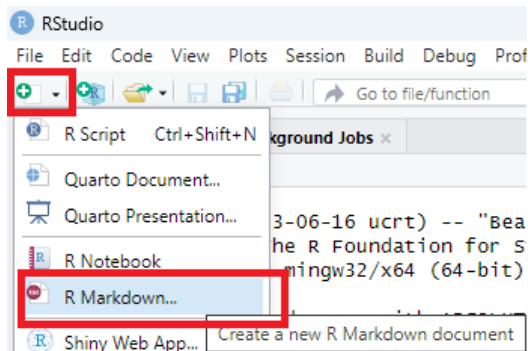
- ▶ Compute basic calculator computations in R.
- ▶ Understand the order of operations in R.
- ▶ Store objects with the assignment operator (`<-`).
- ▶ Understand how to view and remove objects from the workspace.

When done, you'll have to knit your RMarkdown File and make sure you have a .pdf copy to submit.

- ▶ This .pdf will be in preparation for your Homework 1 assignment.

The Markdown

Make sure to open RStudio and create a new R Markdown File. Make sure your name, course, and date is on it.



The R “Chunks”

You will need to add “chunks” into your statement in order to program your R code and run it.



Working directory time!

Make sure you can set your working directory, especially to import datasets into R.

Using the techniques from lecture, create an R Chunk to

- (1) set your working directory, and
- (2) use the `read.csv` command in class to upload the new dataset provided: `ShoeSizeM.csv`. Name the file **SSM**. We'll discuss this concept more in Section 2.10.
- (3) Confirm you imported everything correctly by getting the *summary* values of this variable, using the command line `summary(SSM)`. Note what got added in your **Environment** window!

Calculator Computations

Using Basic Arithmetic to Express Longer Formulas

In lecture, you were guided through operations and the ordering of them. We can also use these operations to computer longer formulas

Say you have a large number of observations (often $n \geq 30$) with sample mean $\bar{x}$ and sample standard deviation s .

Then, the *Central Limit Theorem* suggests that, given $\bar{x}$, the t-test statistic for inference on the mean when $H_0 : \mu = \mu_0$ follows a t-distribution with $(n - 1)$ degrees of freedom (df). It is also written as $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$.

So, if were a given a specific problem, can we compute the test statistic, given our information?

Practice Example & Pseudocoding the Steps

Suppose a labor union wishes to test whether the mean number of hours per month a union member is absent from work differs from $\mu_0 = 12$ hours (1.5 days). The union decides to sample $n = 320$ of its members at random and monitor their working time for 1 month. At the end of the month, the total number of hours absent from work is recorded for each employee. If the mean of the sample is $\bar{x} = 9.8$ hours, and standard deviation is $s = 6.4$ hours, compute the t-test statistic associated with this hypothesis test.

Before you try to write the operations, take a step back and provide the **pseudocode** that explains the order of operations associated with computing this test stat. Label these steps in your Markdown file.

Constructing the test statistic.

Now, go ahead and plug in the values provided in the problem to obtain your test statistic, using correct order of operations.

Objects

Check You Object Know-How!

We'll now return to *objects*

- ▶ An **object** is a storage space (or data structure) with an associated name.

Begin with creating three objects, and assigning the correct values to each:

- (1) `my_name` your first name
- (2) `neighbor_name` the first name of your neighbor
- (3) `self_height` the height of you (any unit of measurement okay)

Question: When you assigned values to these objects, did you have to do anything differently with any of them?

Object Assignment; Going Back to Example

Note that objects can be values, vectors, data sets, and so on.

- ▶ The object name goes on the left of the `<-`, which resembles an arrow pointing left to signify assigning the value on the right to the object on the left.
- ▶ For many formulas, you may wish to create a very general formula that doesn't involve specific values, but (essentially) the *objects* we assign values to (like $\bar{x}$).

So, we'll aim to re-do our example above, but now in two chunks.

New Task!

So, we'll aim to re-do our example above, but now in two chunks.

- ▶ In the first chunk, define the following objects with the values associated with the problem above.
 - ▶ $\bar{x}$ (name it `xbar`), μ_0 (name it `mu0`), n , s .
- ▶ Then, in the second chunk, we will call upon these general objects, and now with values associated with them (so, instead of using 9.8, we will write `xbar` instead). Name this formula the object `tstat`.
- ▶ Run the first chunk, then the second chunk. Finally, create a third chunk that prints off the object `tstat`, which is our formula based the objects we defined above.

Why did we do that?

- ▶ Since each value only showed up once, we might have wondered, “was it worth the extra trouble?” **YES**
- ▶ If you want to *change* the values assigned, you can change them in the *objects* stage first and then re-running your formula will update accordingly! With longer code, finding all the values may be trickier, and so defining the objects at the beginning helps with order and program ‘housekeeping’.
- ▶ If you had multiple instances of a value (like the Z-stat for proportion), you would only need to change the value once instead of multiple times.
- ▶ Finally, this procedure is setting us up to create our own *functions*, using the formulas we want to call upon and deciding which values we want to turn into objects (and eventually which ones will be arguments).

That's It for Today!

Rad job, y'all!

All that is left is to work on converting your Markdown file to a pdf file.

- Either do it directly, or convert it to an html and print to a pdf (see lecture slides for directions). **Make sure your name and UCLA Student ID is on the file somewhere.**



To Get Your Markdown Point

- ▶ To get credit for today, upload the either the html or pdf file only (no .Rmd file needed) in the Discussions & Participation Section of the Modules tab in BruinLearn. Go to Discussion 10/2 and upload there to get your 1 point (Exit Slip).

Enjoy the weekend!